

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech CSE

Semester: V

Name of the Course: Security and Auditing

Course Code: TCS595

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	What is Social Engineering? Discuss different social engineering techniques and methods attackers use to exploit human psychology.	CO1
(b)	Explain Spyware, Adware, and Rogue Software. How do these impact system security?	CO1
(c)	Describe White Hat Search Engine Optimization (SEO) attacks. How do attackers use SEO for cybercrime?	CO1
Q2		
(a)	What are Cyberbullies and Cyber Stalkers? Discuss psychological and digital impact on victims.	CO2
(b)	A small organization is frequently targeted by phishing and fake login pages. As a security auditor, design a step-by-step phishing mitigation plan.	CO2
(c)	Discuss the role of Security Policy in an organization. What components should be included in an effective policy framework?	CO2
Q3		
(a)	Write short notes on: a) SOX compliance b) ISO 27001 standard	CO4
(b)	A company wants to ensure compliance with GDPR and HIPAA. As a consultant, propose a compliance strategy roadmap for the organization.	CO3
(c)	Discuss biometric authentication. What challenges arise in storing and securing biometric data?	CO3
Q4		
(a)	Differentiate between symmetric and asymmetric encryption with examples.	CO5
(b)	Explain SSL/TLS encryption. What happens during the TLS handshake process?	CO5
(c)	Suggest an authentication and encryption model for securing a university online examination system using MFA and SSL/TLS	CO4
Q5		
(a)	Explain Database Activity Monitoring (DAM). Why is it essential for enterprise database security?	CO5
(b)	Describe the stages of the database lifecycle. Explain the security concerns at each stage.	CO6
(c)	Discuss Database Backup, Recovery, and Disaster Recovery Planning. Why are these critical for business continuity?	CO6

